



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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https://github.com/RobertMatei35/DataScienceCert/tree/main/10_AppliedDataScienceCapstone



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodologies

- The project followed an end-to-end data science workflow:
- Collected SpaceX launch data using APIs and web scraping.
- Cleaned and transformed the data, including payload, orbit, launch site, booster, and landing outcome variables.
- Used SQL, Pandas, Matplotlib, and Seaborn for exploratory analysis.
- Created interactive Folium maps and a Plotly Dash dashboard to analyze launch locations, outcomes, and payload ranges.
- Standardized features, split the data into training and test sets, and tuned Logistic Regression, SVM, Decision Tree, and KNN models using 10-fold GridSearchCV.

Results

- Landing success generally improved over time and with greater flight experience.
- Launch sites were located near coastlines, mainly in Florida and California.
- Payload mass, orbit type, launch site, and booster generation influenced landing success.
- The models achieved approximately 83.3% test accuracy. The Decision Tree had the strongest cross-validation performance, although the small test set means the results should be interpreted cautiously.

Introduction

- Space industry is a continuous growing industry with the increase of the technological advancements. This means an increase of the competition in this sector where every cost reduction measure offers an edge on the competition. Currently SpaceX advertises that the cost of a Falcon 9 rocket launch is around 62 million dollars while competition has a cost upward of 165 million each.
- This cost advantage is due to the reusability of the first stage of the Falcon 9 rocket. The project's goal is to determine the cost of a launch based on the successfully landing of the first stage. To be able to achieve the goal we will collect, process and model the data to determine the chances of a successful landing of a future launch based on some characteristics.



Section 1

Methodology

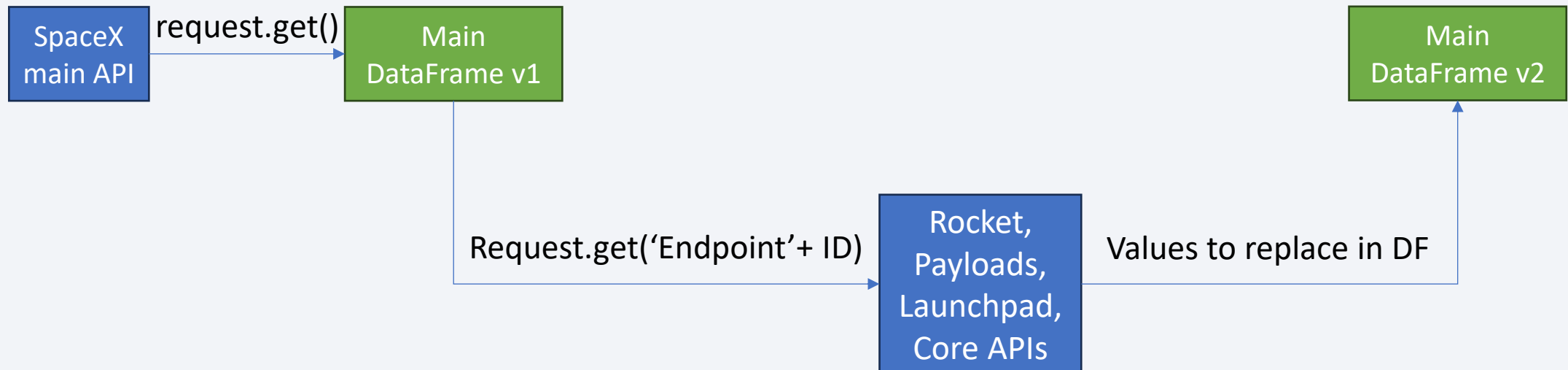
Methodology

Executive Summary

- Data collection methodology:
 - Data was collected from SpaceX API and SpaceX wiki page
- Perform data wrangling
 - Data processing and Training Label determination was performed in this step
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Different ML models(Logistic Regression, SVM, Decision Trees and KNN) were created, tuned and validated to observe the best performing model

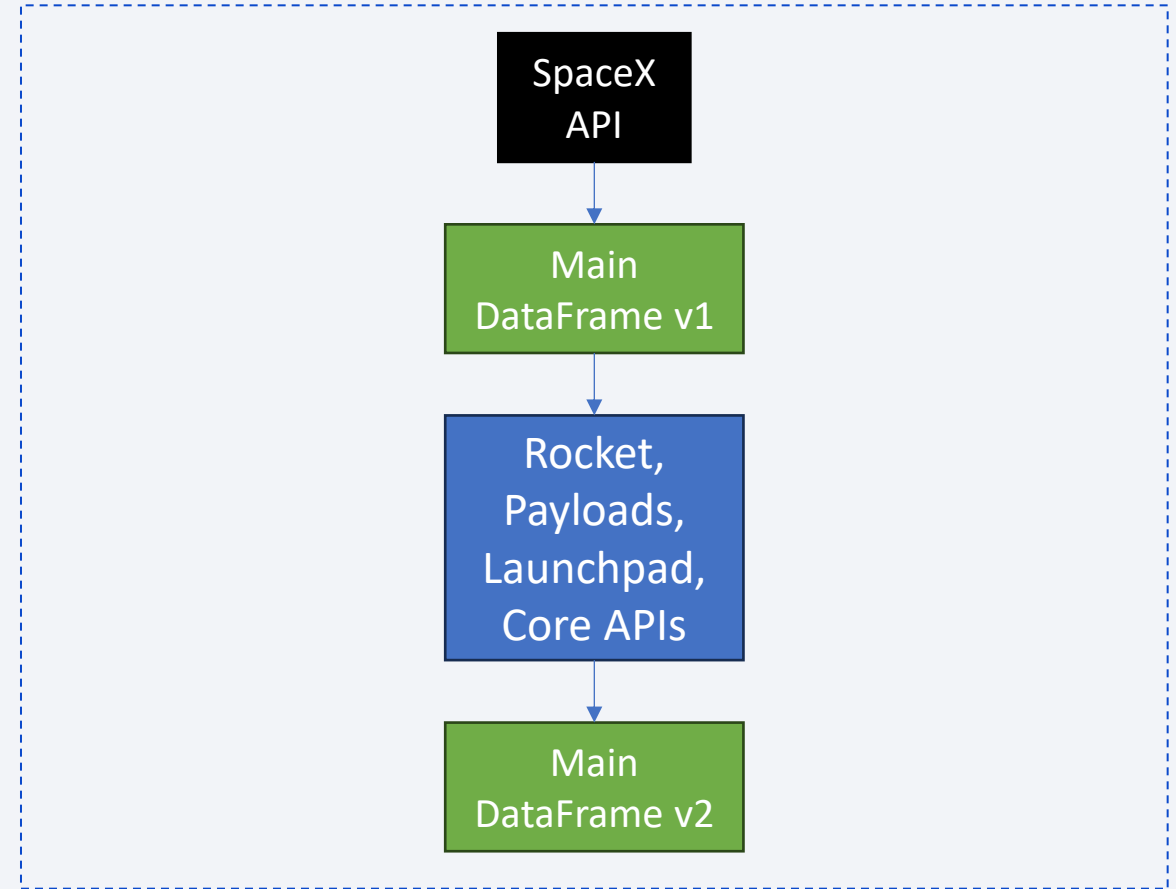
Data Collection

- Dataset was collected from the SpaceX API using Requests package in python
- The main dataset contained database ID columns for: rocket, payloads, launchpad and id. To collect the actual values an API endpoint had to be accessed for each of the variables above.
- After getting the values for the ID specific functions are used to process the data in the main dataset



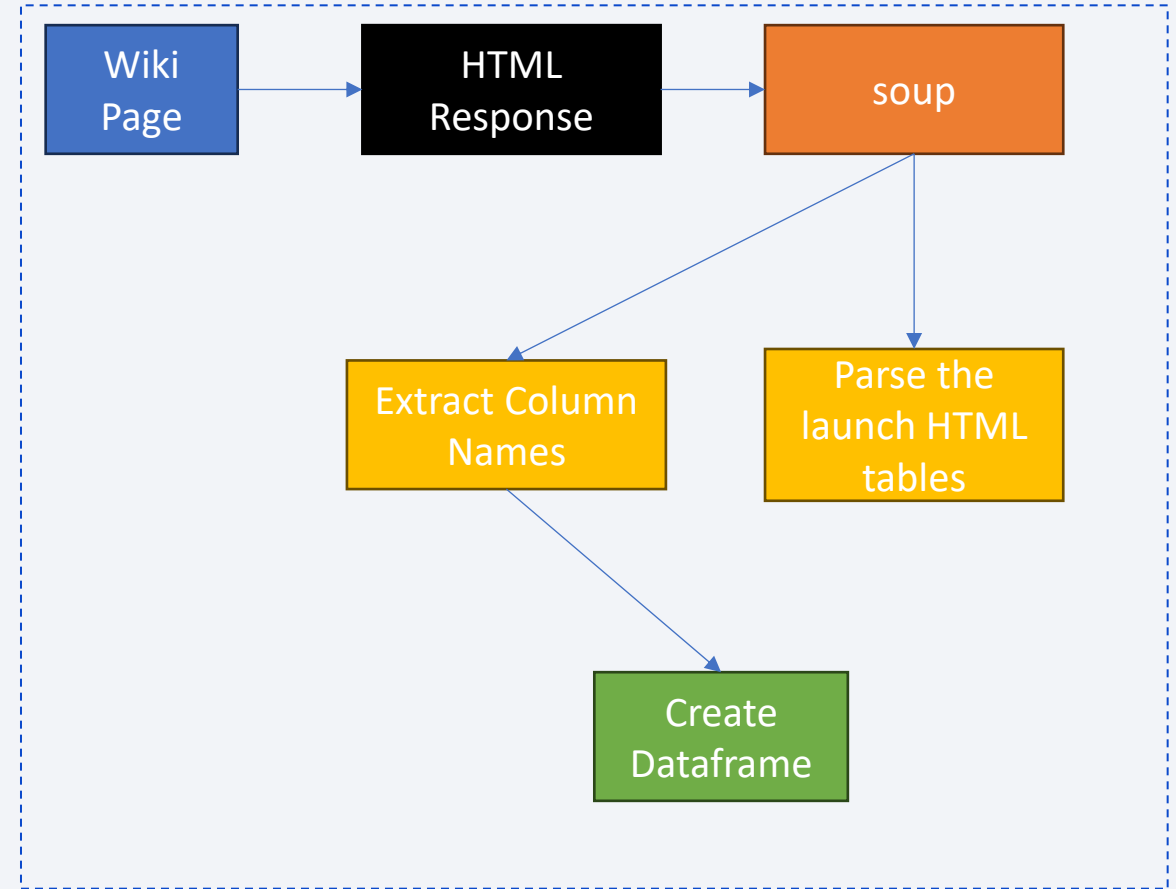
Data Collection – SpaceX API

- Objectives at this step were:
 - Create a request to the SpaceX API
 - Process the retrieved data
- Data was retrieved from a main Endpoint
- Rocket, Launchpad, Payload and Core data contained databased id keys
- Additional endpoints were used to get actual values instead of keys
- [DataScienceCert/10_AppliedDataScienceCapstone/Module 1 - Introduction/jupyter-labs-spacex-data-collection-api-v2.ipynb](#) at [main · RobertMatei35/DataScienceCert](#)



Data Collection - Scraping

- Similar objective as data collection – collect falcon 9 historical launch records from Wikipedia page
- Methodology is presented in the adjacent flowchart
- [DataScienceCert/10_AppliedDataScienceCapstone/Module 1 - Introduction/jupyter-labs-webscraping.ipynb](#) at main · [RobertMatei35/DataScienceCert](#)



Data Wrangling

- To process the data from its raw form and make it more usable during the analysis key processes were made:
 - Familiarize with the data
 - Perform calculations based on the data
 - Create new label column useful for the future models
- [DataScienceCert/10 AppliedDataScienceCapstone/Module 1 - Introduction/labs-jupyter-spacex-Data wrangling-v2.ipynb at main · RobertMatei35/DataScienceCert](#)

EDA with Data Visualization

- Multiple charts were plotted at this step:
 - Scatter plots – present relation between pairs of variables (e.g. Payload and Launch Site)
 - Categorical plots – display relation between pairs of variables like a scatter plot, but it is used for categorical columns (e.g. Payload and Launch site)
 - Bar charts – used to display data grouped in categories. In the project it was used to highlight the success rate for each orbit type
 - Line plots – used to observe the evolution in time of a variable. In this case the success rate evolution is presented
- [DataScienceCert/10 AppliedDataScienceCapstone/Module 2 - Exploratory Data Analysis/jupyter-labs-eda-dataviz-v2.ipynb at main · RobertMatei35/DataScienceCert](#)

EDA with SQL

- SQL Queries performed:
 - Display the names of launch sites with various conditions
 - Display payload mass of different rockets
 - List the date of the first successful landing in a specific ground pad
 - Create subqueries to present booster_versions that have carried maximum payload mass
 - Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- [DataScienceCert/10 AppliedDataScienceCapstone/Module 2 - Exploratory Data Analysis/jupyter-labs-eda-sql-coursera sqlite.ipynb at main · RobertMatei35/DataScienceCert](#)

Build an Interactive Map with Folium

- Multiple map objects as markers, circles and lines were created to:
 - Create visualization of the launch sites
 - Mark success and failed launches for each site on the map
 - Calculate and visualize distances between launch site to its proximities
- Map objects usage:
 - Circles are used to highlight the coordinates of the launch sites
 - Markers are used for annotation that give context to the circles. Moreover, markers are used to mark success/failed launches
 - Lines connect the launch sites to proximity infrastructure
- [DataScienceCert/10_AppliedDataScienceCapstone/Module 3 - Interactive Visual Analytics and Dashboard/lab-jupyter-launch-site-location-v2.ipynb at main · RobertMatei35/DataScienceCert](#)

Build a Dashboard with Plotly Dash

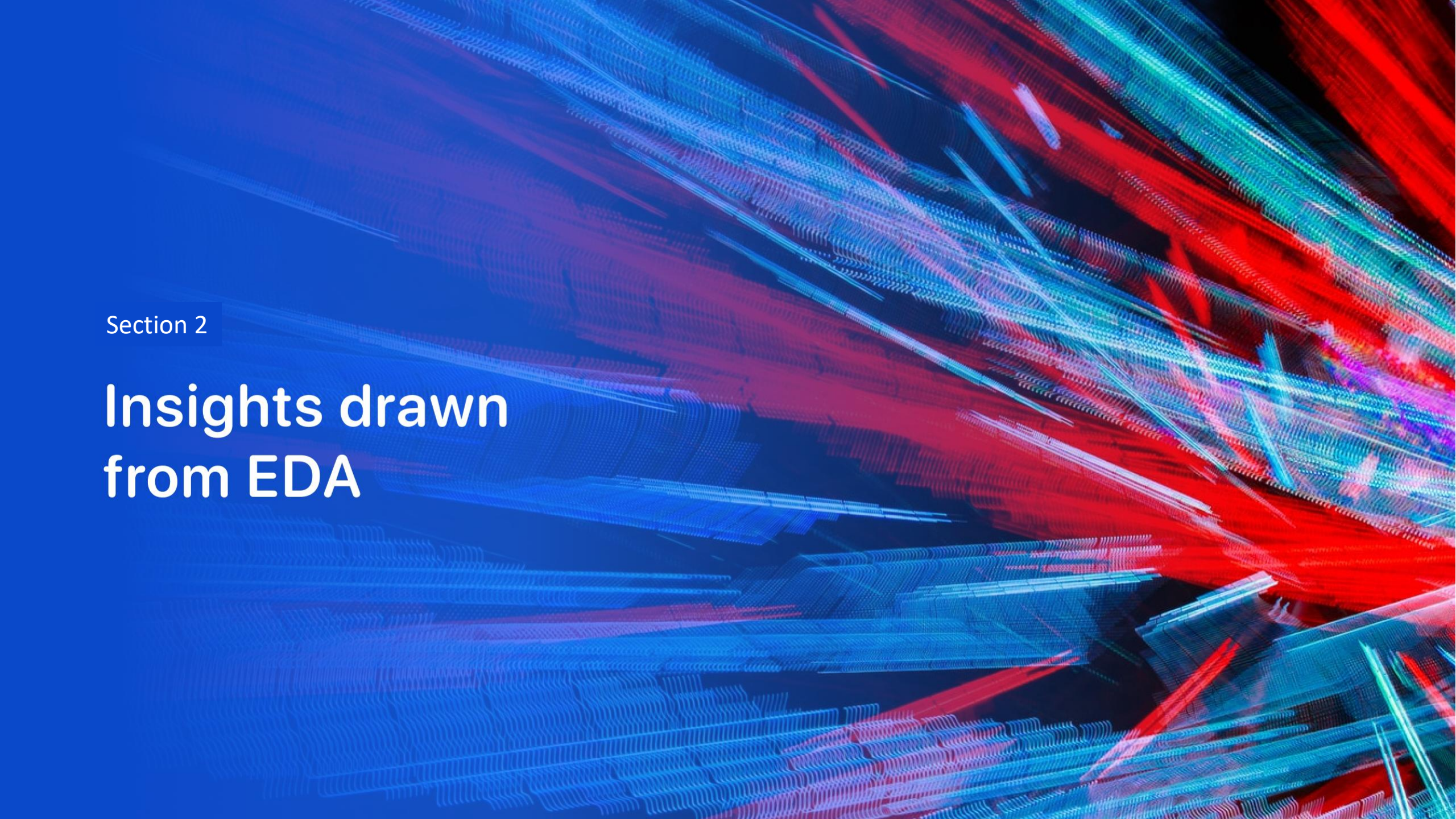
- The Dashboard contains:
 - Charts/Plots:
 - Pie Chart – illustrates launch successes based on launch sites
 - Scatter Plot – Shows the class(success/failure) of a launch based on the payload mass. Each point is colored depending on its booster version category
 - Interactions
 - Dropdown – It offers the user the possibility of selecting which launch site is of interest for him
 - Slider – Lets user reduce or increase the payload mass range of the flights that are displayed in the scatter plot
- [DataScienceCert/10_AppliedDataScienceCapstone/Module 3 - Interactive Visual Analytics and Dashboard/spacex-dash-app.py at main · RobertMatei35/DataScienceCert](#)

Predictive Analysis (Classification)

- To obtain the best classification model range of models were trained and tested:
 - Logistic Regression
 - Support Vector Machine
 - Decision Tree
 - K-Nearest Neighbors
- Steps performed in predictive analysis:
 - Create X and y variables
 - Standardize X data
 - Create train and test datasets
 - Create machine learning model, hyperparameters and a GridSearchCV object
 - Fit the model, then analyze the results
- [DataScienceCert/10_AppliedDataScienceCapstone/Module 4 - Machine Learning/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb at main · RobertMatei35/DataScienceCert](#)

Results

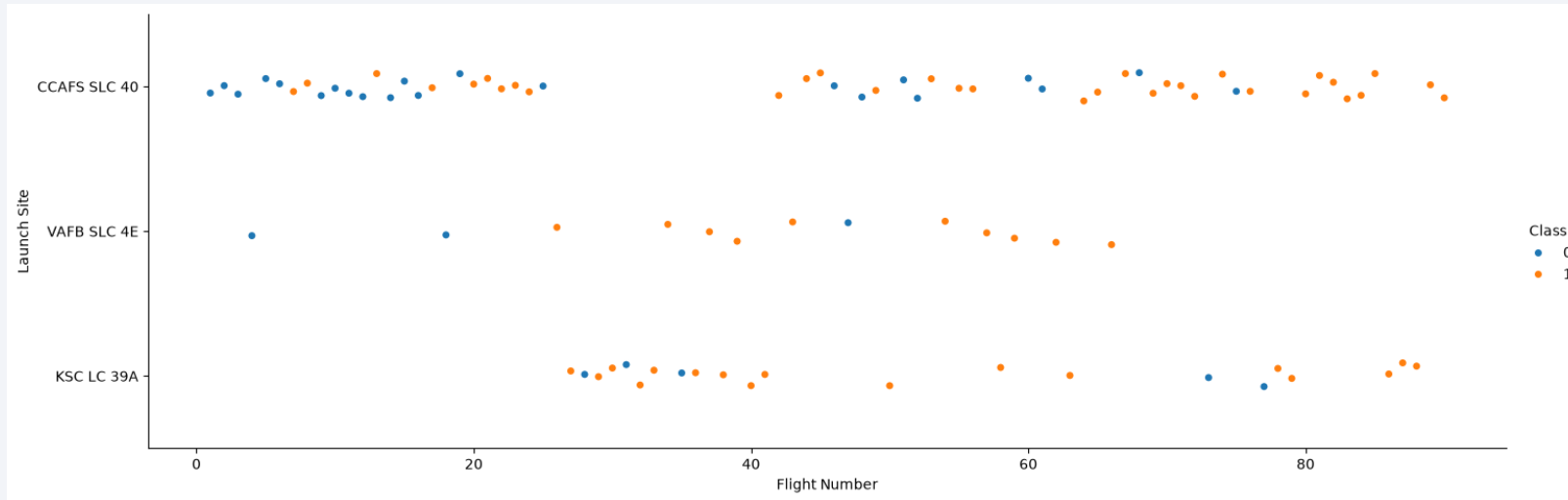
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

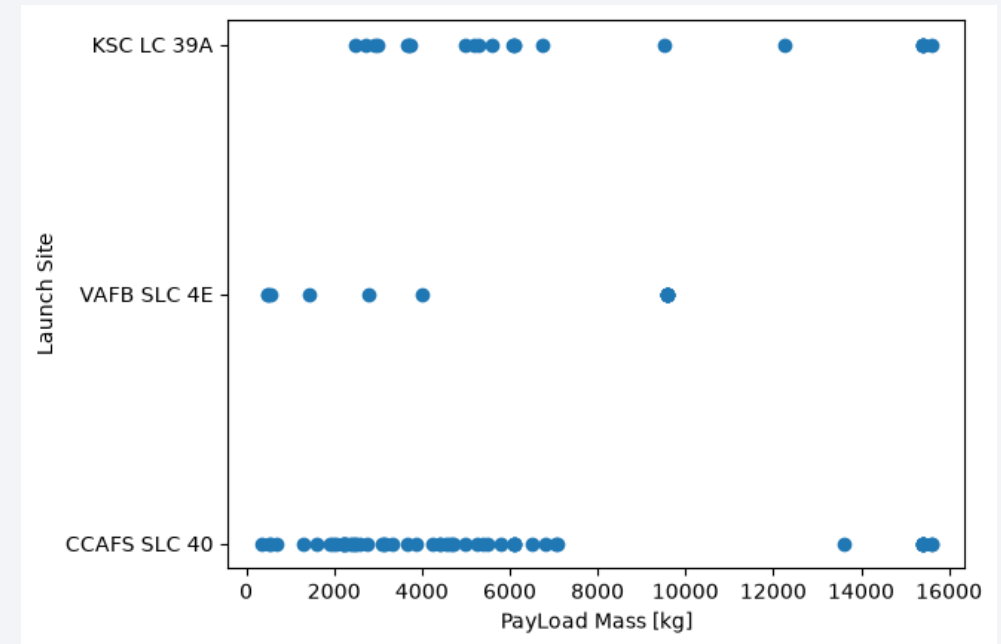
Insights drawn from EDA

Flight Number vs. Launch Site



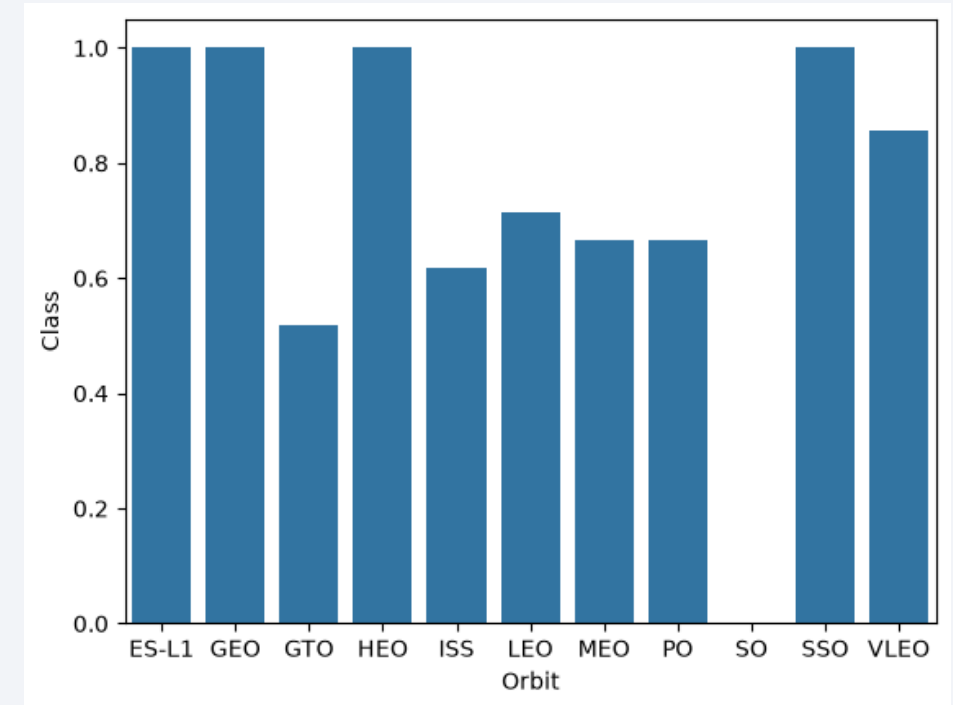
Payload vs. Launch Site

- The KSC launch site has the most distributed payload mass
- The VAFB launch site has the least entries and they are concentrated towards the lower payload mass
- CCAFS SLC have loads of entries between 0 and 8000 kgs with few entries above 12000 kgs

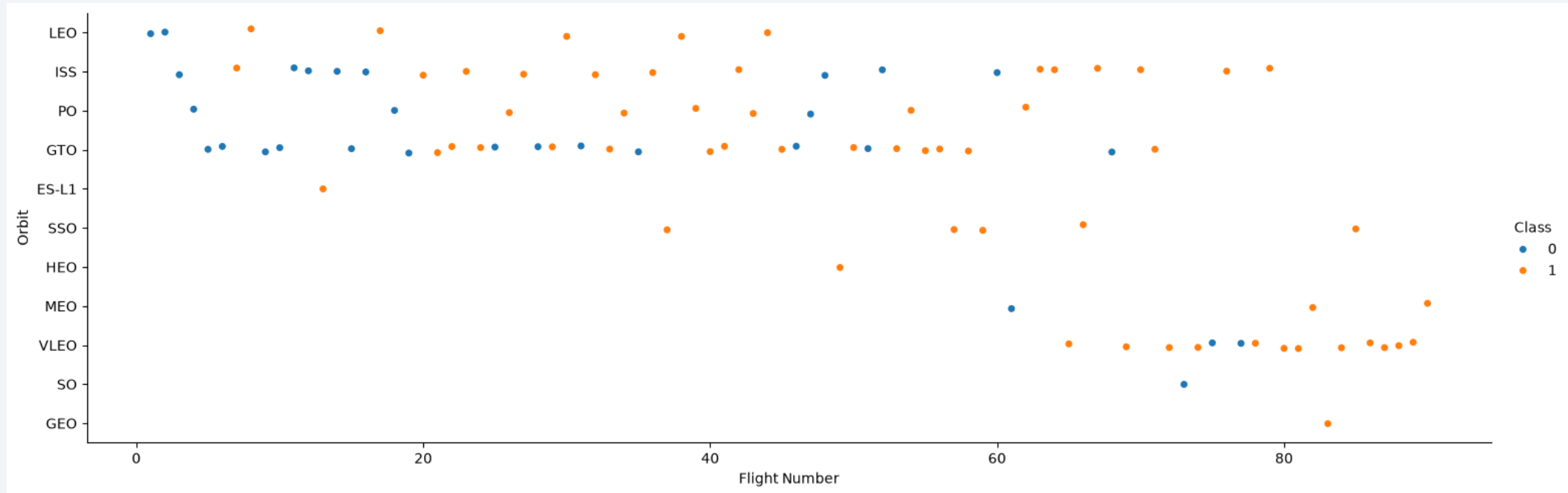


Success Rate vs. Orbit Type

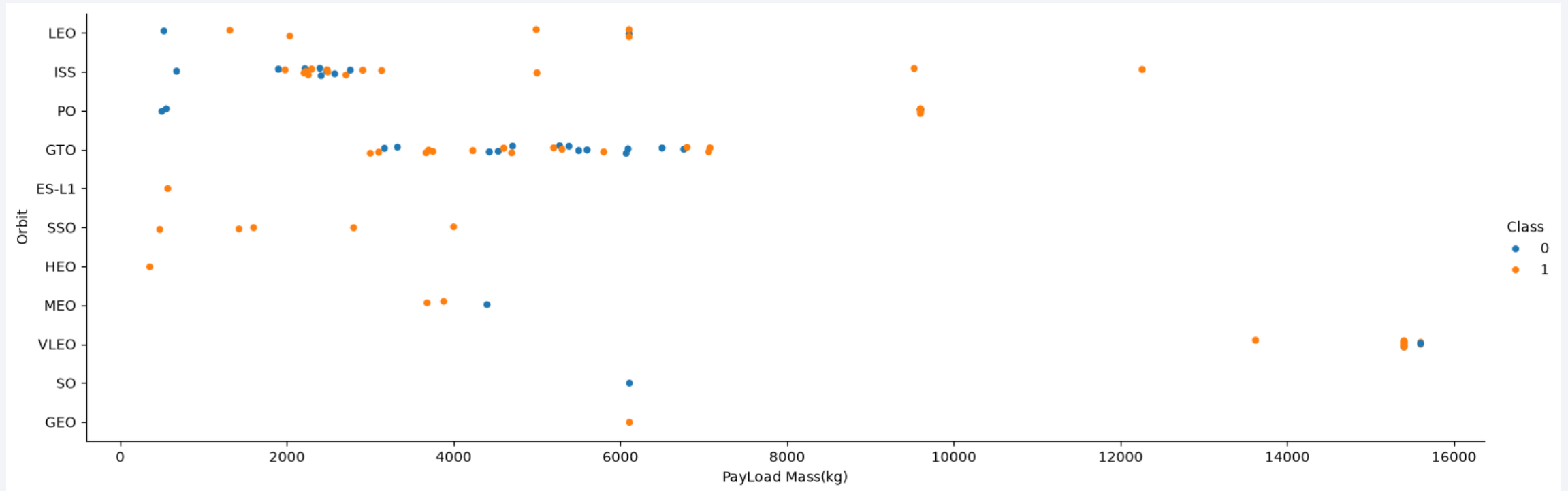
- There are few orbits that have 100% success rate
- The least successful rate is the SO orbit which is 0%
- ISS, LEO, MEO, PO have similar success rates
- GTO orbit is the least successful orbit which is not 0%



Flight Number vs. Orbit Type



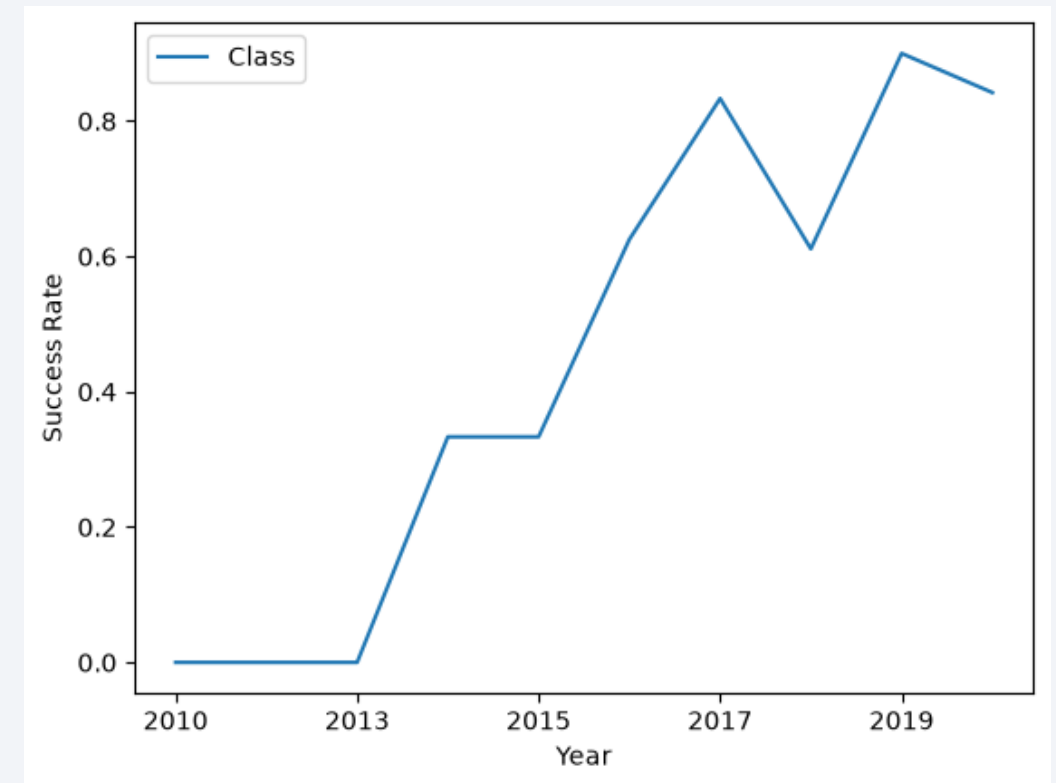
Payload vs. Orbit Type



- With heavy payloads the successful landing rate are increased for Polar, LEO and ISS.
- However, for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are both there here.

Launch Success Yearly Trend

- The success rate is increasing with the year
- In the first years, the success rate was 0 and with the passing of the years the success increased
- The behavior is in accordance with technological development



All Launch Site Names

- All launch site names is presented in the list below:
 - CCAFS LC-40
 - VAFB SLC-4E
 - KSC LC-39A
 - CCAFS SLC-40
- The SQL query used is: %sql select distinct "Launch_Site" from SPACEXTBL

Launch Site Names Begin with 'CCA'

- The first 5 records where launch sites begin with 'CCA' are shown in the screenshot below

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The outcome of these missions is a success, but the landing is either failed or no attempt at landing
- The query used is: %sql select * from SPACEXTBL WHERE "Launch_Site" like 'CCA%' limit 5
- The query selects all variables of the database where the Launch Site begins with CCA
- The records queried are the first 5 in the database without any explicit order

Total Payload Mass

- The total payload mass carried by NASA booster is 45596 kg

```
SUM(PAYLOAD_MASS_KG_)
45596
```

- The query used is: %sql select SUM(PAYLOAD_MASS_KG_) from SPACEXTBL where Customer = 'NASA (CRS)'
- The query creates a sum of all entries of the Payload Mass kg where the customer variable value is NASA (CRS)

Average Payload Mass by F9 v1.1

- The average value of the payload mass of the F9 v1.1 booster is 2928.4 kg

AVG(PAYLOAD_MASS_KG_)
2928.4

- Query used for this calculation is: %sql select AVG(PAYLOAD_MASS__KG_) from SPACEXTBL where "Booster_Version" = 'F9 v1.1'
- The query calculates mean payload mass kg only in the rows where the Booster_Version variable is equal to F9 v1.1

First Successful Ground Landing Date

- The first successful landing on a ground pad was 2015-12-22

MIN(Date)
2015-12-22

- The query used was: %sql select MIN("Date") from SPACEXTBL where "Landing_Outcome" = 'Success (ground pad)'
- The query selects the minimum value for date and checks that the Landing_Outcome variable has the value Success (ground pad)

Successful Drone Ship Landing with Payload between 4000 and 6000

- The boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:
 - F9 FT B1022
 - F9 FT B1026
 - F9 FT B1021.2
 - F9 FT B1031.2
- To determine which boosters respect the condition the following query was used:

```
%sql
select "Booster_Version" from SPACEXTBL where "Landing_Outcome" = 'Success (drone
ship)' and PAYLOAD_MASS__KG_ between 4000 and 6000
```
- The query selects booster versions that respect two conditions:
 - Landing Outcome to be a success
 - Payload Mass to be between 4000 and 6000 kgs

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The successful missions happen at a rate of 100x than failed missions. The results say that there are 100 successful missions and only 1 failed.

SUCCESSFUL MISSION
100

FAILED MISSION
1

- Two queries were used to determine the information needed for this task:
 - %sql select count(*) as 'SUCCESSFUL MISSION' from SPACEXTBL where "Mission_Outcome" like 'Success%'
 - %sql select count(*) as 'FAILED MISSION' from SPACEXTBL where "Mission_Outcome" like 'Failure%'

Boosters Carried Maximum Payload

- There are multiple boosters which have carried the maximum payload mass.
- The list of the boosters is shown in the screenshot presented in this slide
- The query used to obtain the results is: %sql select "Booster_Version" from SPACEXTBL where PAYLOAD_MASS__KG_ = (select MAX(PAYLOAD_MASS__KG_) from SPACEXTBL)
- The query uses a subquery to determine the maximum value of the payload mass. The result is then compared in the where clause of the main query

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- The failed landing outcomes in drone ship for year 2015 are illustrated in the screenshot of this slide. Only two failures happened searching by this criteria

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- The query used is: %sql select substr(Date,6,2) as 'Month', "Landing_Outcome", "Booster_Version", "Launch_Site" from SPACEXTBL where "Landing_Outcome" = 'Failure (drone ship)' and substr(Date,0,5)='2015'
- To obtain the month and year needed in this query SUBSTR(string, start position, length) function was used

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The rank of landing outcomes between 2010-06-04 and 2017-03-20 is presented in the screenshot.
- The most frequent landing outcome is the no attempt. Then all the successes for a given platform are greater or at least equal to the failures on the same platform
- The query used is: %sql select "Landing_Outcome", Count("Landing_Outcome") as 'Count' from SPACEXTBL where Date between '2010-06-04' and '2017-03-20' group by "Landing_Outcome" order by "Count" desc

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue rectangle on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible, separating the dark surface from the deep blue of the atmosphere and the blackness of space.

Section 3

Launch Sites Proximities Analysis

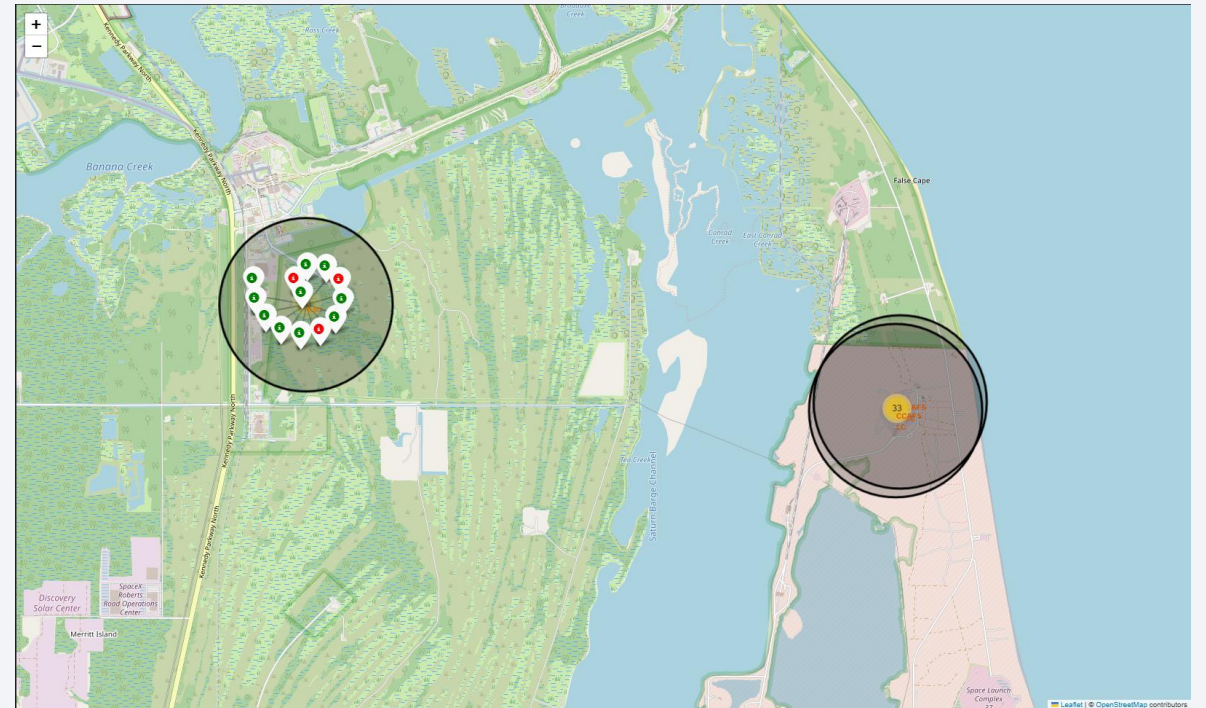
Launch Sites Overview

- The map presents the launch sites positioning on the USA map
- The launch sites are divided between eastern and western coasts
- There is only one launch site on the west coast, namely VAFB
- All the other sites are close to each other in Florida



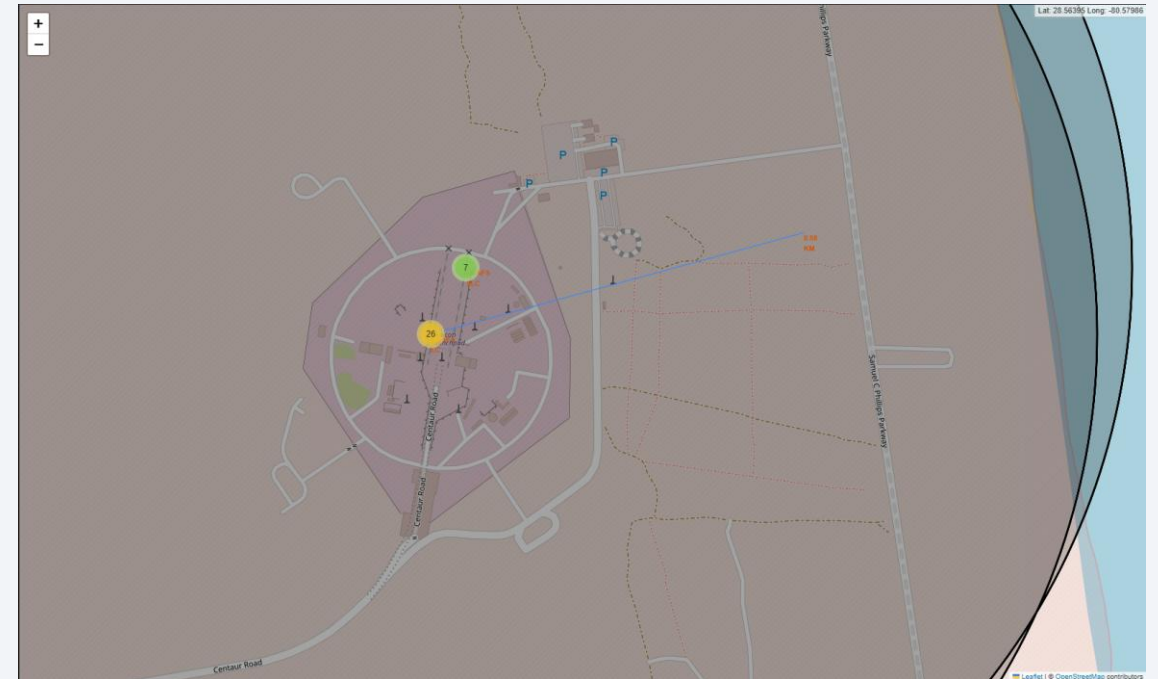
Launches Outcome Visualization

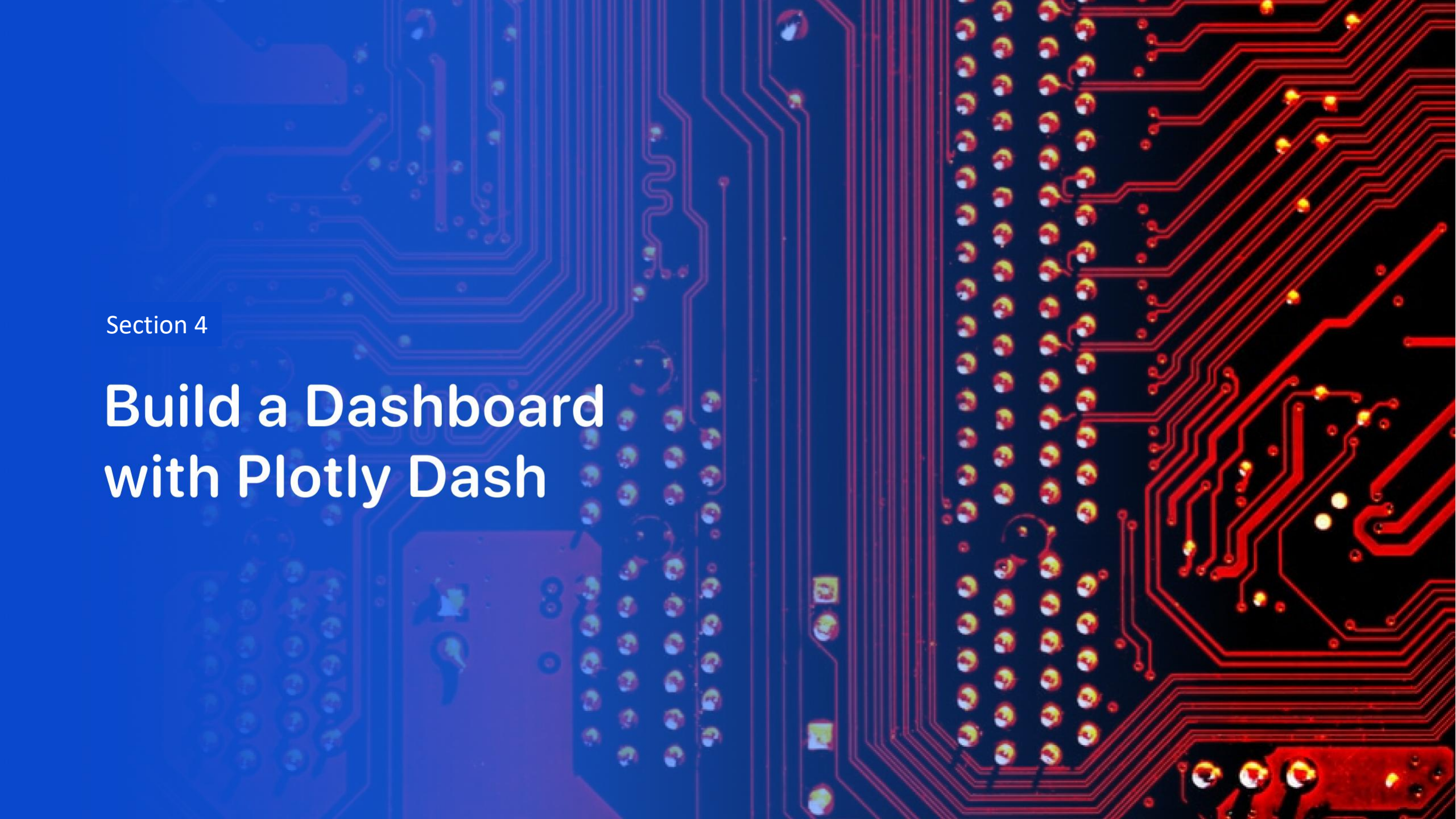
- The illustration present the outcome of the launches from the KSC site
- There are 10 successful outcomes to only 3 failures
- In the close neighborhood there is the CCAFS launch sites



CCAFS launch sites proximities

- The map presents the distance between CCAFS site launches to the closest highway
- The blue line represents the straight line from the site to the highway
- The orange text represents the distance: 0.58 km

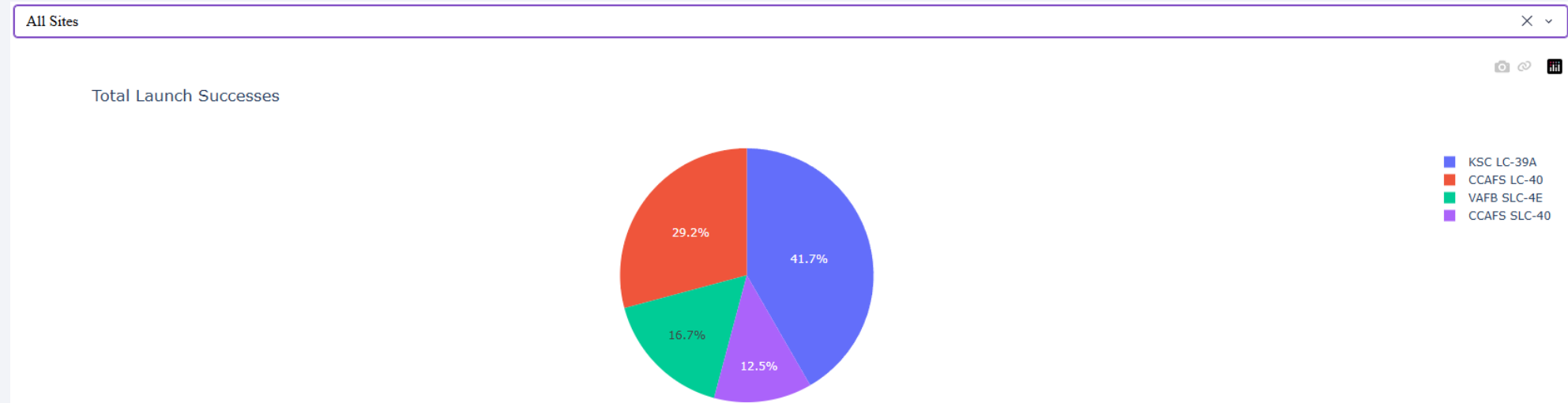




Section 4

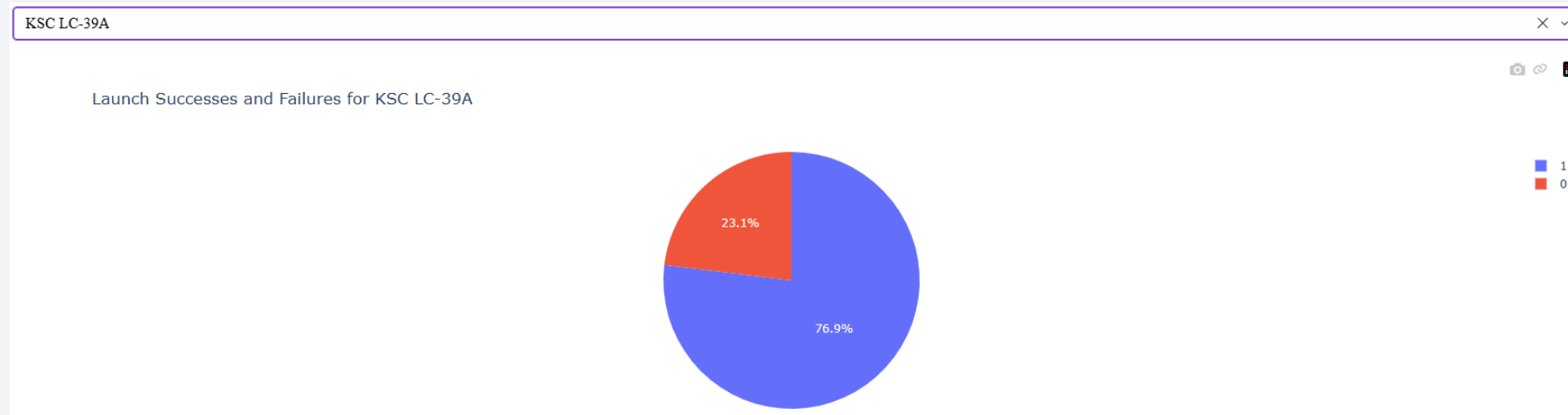
Build a Dashboard with Plotly Dash

Launch Successes of all Sites



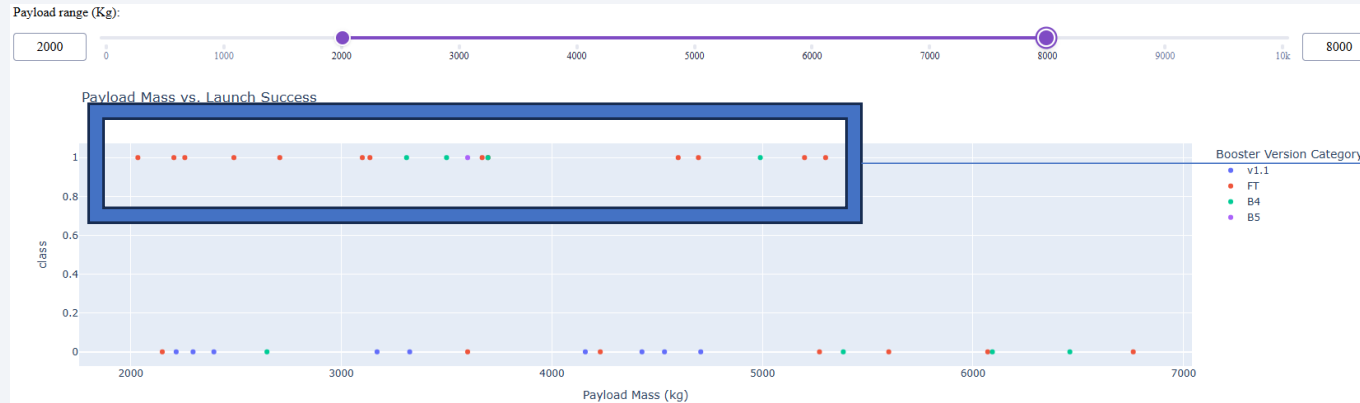
- The total successes from all sites is displayed using a pie chart
- The most successful site by far is the KSC LC-39A launch site.
- The second is the CCAFS LC-40
- The CCAFS SLC-40 and VAFB SLC-4E have a lower share between 12 and 17%

Most Successful site breakdown



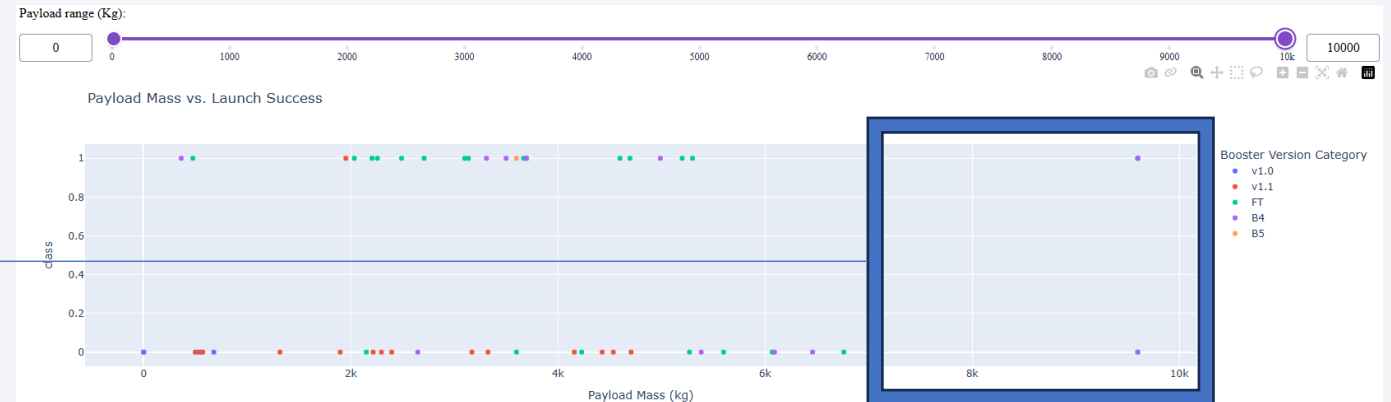
- The piechart shows the distribution between failed and successful landing outcomes for KSC LC-39A
- This site has the highest launch success ratio
- Blue part represents successful landings and red part represents failed landings

Success depending on Payload and Booster Version



Lower ranges provide multiple successful launches

Higher payload ranges exhibit an increase of failed launches

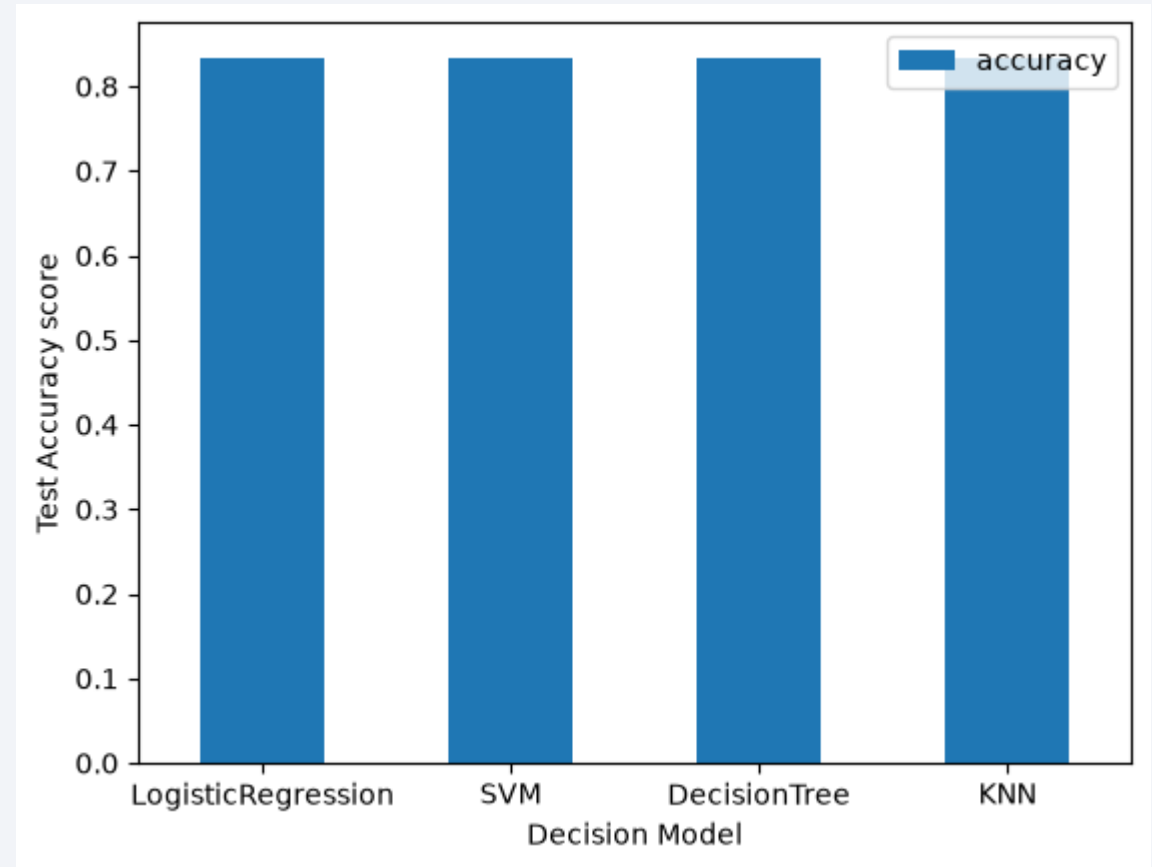


Section 5

Predictive Analysis (Classification)

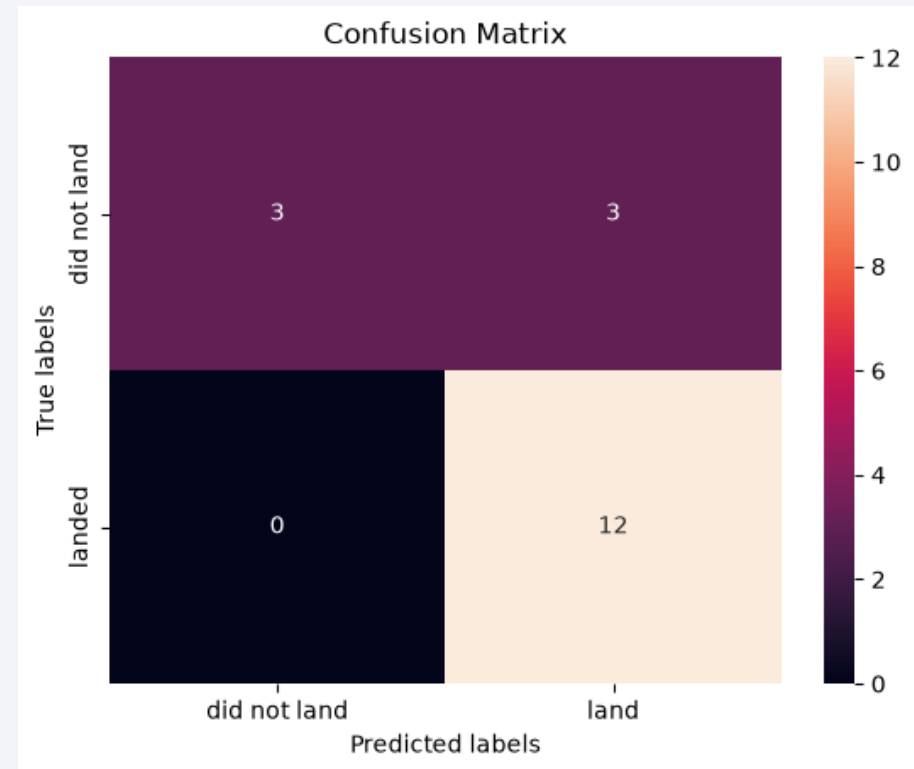
Classification Accuracy

- The accuracy for all four decision models is equal at: 0.83
- The accuracy measured on training data is different.
- The standout in training data is the decision tree model at 0.89



Confusion Matrix

- All models performed the same on testing data
- The model performs perfectly when it needs to predicts True Positives (100% score)
- The model has issues when it is required to determine True Negatives
- The score between True Negatives and False Negatives is 50-50%
- The performance of the model when the rocket did not land is not inspiring confidence



Conclusions

- The main conclusion drawn after this project is that the Falcon 9 landing success improved over time, with stronger results associated with later flights
- Payload and Orbit influenced the landing results, with heavier and more demanding missions presenting greater landing difficulty
- Launch sites are present on the coast, mainly Florida and California
- Machine Learning models achieved 83.3% test accuracy. The decision tree had the best cross validation score.
- Test set contained only 18 samples meaning the results are not yet statistically robust

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

